

Devran V. Turson

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EXPERIENCE

IMADE3D Centreville, VA
Independent Contractor – Embedded Systems Software May 2026 - July 2026

- Developed and maintained the JellySTEM MakeCode extension (pxt-jellystem) for the BBC micro:bit V2, implementing block-based APIs for OLED displays, NeoPixel LEDs, ultrasonic sensors, and motor control
- Engineered a custom SSD1306 framebuffer renderer in TypeScript to enable pixel-perfect filled shapes, bypassing limitations of the underlying display library
- Built and structured a hardware documentation repository covering verified components, wiring guides, MakeCode examples, and ready-to-flash .hex test programs

James Madison University Harrisonburg, VA
Haynes Leader / Teaching Assistant October 2025 - Present

- Represent the Haynes program at Open House and CHOICES recruitment events; follow up with prospective students
- Create and maintain contact lists for current and former Haynes scholars to support program communications
- Serve as a Learning Assistant for Math 167 (Haynes Seminar), supporting student learning during Monday/Wednesday sessions
- Onboard incoming Haynes Scholars via email, coordinating course registration, housing, and faculty advisor details
- Coordinate social events and communications across the Haynes community

JEM Lab Technician March 2025 - Present

- Operated and maintained FDM and resin 3D printers (Ultimaker S3, Form 3+) using Cura, Preform, and OpenSCAD; designed STL files and handled routine troubleshooting
- Produced 3D printed mathematical models for STEM outreach and educational demonstrations

First Year Student Researcher – FYRE Program (Twoville) January 2025 – May 2025

- Collaborated on Twoville, a bidirectional code-and-mouse programming language for 2D/3D digital fabrication

PROJECTS

Routine Clicker – Full Stack July 2026

- Built an ESP32 hardware button that syncs habit tracking data to Firebase Realtime Database in real time
- Developed a JavaScript/HTML/CSS web dashboard featuring a GitHub style heatmap calendar and OLED progress bar for commit tracking
- Implemented C++ firmware on the ESP32 handling Wi-Fi connectivity and Firebase REST API calls for real time data sync

pxt-jellystem – MakeCode Extension May 2026

- Built a MakeCode extension for the BBC micro:bit V2 enabling students to control OLED displays, NeoPixel LEDs, servo motors, and sensors through a beginner-friendly block interface
- Designed a unified hardware abstraction layer in TypeScript so students can swap between I²C and PWM servo setups without changing their code
- Restructured the extension's block menu for student readability, matching group names to physical hardware labels for intuitive navigation

SKILLS

Languages	Tools
Python, Java, C, C++, TypeScript, JavaScript, HTML, CSS, OpenSCAD, Markdown, NetLogo, Twoville	Git, GitHub, VS Code, Linux/Unix, Node.js, REST APIs, Firebase, Arduino IDE, BBC micro:bit, MakeCode, GCC, GNU Make, Figma, Cura, Preform

EDUCATION

James Madison University Bachelor of Science
Computer Science Harrisonburg, VA • 2024 - 2027 (December)